

# BAVARIA COP 1809

## *Common Operating Picture*

### User Guide

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Flight of the Eagle Wargame | U.S. Army War College | 1809 Danube Campaign

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*Version 2.0 — May 2025 | Road Network Edition*

*FOR EDUCATIONAL USE — USAWC PROFESSIONAL MILITARY EDUCATION*

## 1. Overview

Bavaria COP 1809 is a browser-based Common Operating Picture for the Flight of the Eagle wargame at the U.S. Army War College. It tracks unit positions, calculates road-network march distances, visualizes movement orders, monitors contact between opposing forces, and displays the 1809 Order of Battle hierarchy.

The application runs in a web browser. All distance calculations use a hand-built road network matching the period road system shown on the game map — not straight-line distances. This means ETAs and march times match what your umpire calculates.

### 1.1 System Requirements

| Requirement | Details                                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Browser     | Google Chrome or Microsoft Edge                                                                                                            |
| Internet    | Required on first open to cache map tiles (OpenStreetMap)                                                                                  |
| Files       | All four files must be in the same folder: bavaria_cop_leaflet.html, game_map_georef.jpg, alps_map_georef.jpg, rhine_danube_map_georef.jpg |
| Screen      | 1280×720 minimum recommended                                                                                                               |

### 1.2 Opening the Application

- Download all four files to the same folder on your computer
- Do not rename the image files — the app references them by exact filename
- Open Chrome and drag bavaria\_cop\_leaflet.html into the browser
- The map loads centered on the Bavaria 1809 theater

**Note:** If period map layers show a broken image icon, confirm all four files are in the same folder.

## 2. Interface Layout

### 2.1 Header Bar

| Control                                                                                                                                                                          | Function                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Clock Display                                                                                                                                                                    | Current game DTG: DD HHMM APR (e.g. 16 0600 APR)             |
| -1H / -30M                                                                                                                                                                       | Step clock backward                                          |
| +30M / +1H                                                                                                                                                                       | Step clock forward                                           |
| Speed Selector                                                                                                                                                                   | Playback speed: 1×, 5×, 30×, 60×                             |
|  PLAY /  PAUSE | Start or stop automatic clock                                |
| Map Layer Selector                                                                                                                                                               | Toggle between Street Map, Game Map, Alps 1809, Rhine-Danube |

### 2.2 Perspective Toggle

The PLAYING AS bar sets which faction appears as Friendly (blue rectangle) and which appears as Enemy (red diamond).

- FRENCH — French units show as blue rectangles, Austrians as red diamonds
- AUSTRIAN — Austrians show as blue rectangles, French as red diamonds

### 2.3 Map Layers

| Layer                                                                                            | Description                                                         |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
|  Street Map   | OpenStreetMap — modern roads and geography, always accurate         |
|  Game Map     | Pratzen-style Bavaria 1809 tactical map — georeferenced to real GPS |
|  Alps 1809    | Thiers/Dufour Carte des Alpes — strategic theater overview          |
|  Rhine-Danube | Thiers/Dufour Carte No. 29 — Napoleon's western approach axis       |

**Note:** The road network distance calculator works regardless of which map layer is displayed. The map is a visual background only.

### 3. Road Network Distance Calculator

This is the most important feature in Version 2.0. All distance and ETA calculations now use a hand-built road network based on the period road system shown on the game map. The app uses Dijkstra's shortest-path algorithm to find the best route between any two towns.

#### 3.1 Distance Modes

In the MOVE tab, toggle between two distance modes:

| Mode                                                                                            | Description                                                 | Use When                                        |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------|
|  ROAD          | Shortest path through the road network — matches game rules | Always — this is the default and correct mode   |
|  AS CROW FLIES | Straight-line distance between GPS coordinates              | Comparison only — will underestimate march time |

#### 3.2 Verified Key Road Distances

These distances have been verified against John H. Gill's monograph and match the game umpire reference:

| Route                   | Road Distance | Via                                       |
|-------------------------|---------------|-------------------------------------------|
| Straubing → Regensburg  | 44 km         | Wörth an der Donau → Barbing              |
| Regensburg → Alkofen    | 14 km         | Direct                                    |
| Regensburg → Abensberg  | 32 km         | Kelheim                                   |
| Regensburg → Landshut   | 70 km         | Eggmühl → Schierling → Mallersdorf        |
| Ingolstadt → Landshut   | 64 km         | Mainburg → Pfeffenhausen                  |
| Straubing → Passau      | 93 km         | Geiselhöring → Plattling → Deggendorf     |
| Donauwörth → Regensburg | 102 km        | Neuburg → Ingolstadt → Kelheim            |
| Landshut → Braunau      | 137 km        | Freising → Erding → Mühldorf → Burghausen |

#### 3.3 Unit March Speeds

| Unit Type                  | Speed   | Examples                          |
|----------------------------|---------|-----------------------------------|
| Army/Corps HQ courier      | 8 km/hr | Napoleon, Berthier                |
| French/Austrian Cavalry HQ | 5 km/hr | Davout HQ, Vandamme HQ            |
| French Cavalry             | 5 km/hr | Montbrun, Saint-Sulpice, Nansouty |
| French Line Infantry       | 4 km/hr | Morand, Friant, Gudin, Demont     |
| Austrian Cavalry           | 4 km/hr | FML von Kienmayer, Liechtenstein  |
| Austrian Infantry          | 3 km/hr | Hohenzollern, Rosenberg, Hiller   |

**Note:** No terrain modifiers or fatigue are applied by the COP. Apply those manually per the umpire rules.

## 4. PLACE Tab — Adding and Removing Units

### 4.1 Place Unit via Dropdown

- Select FRENCH or AUSTRIAN faction
- Pick unit and town from dropdowns
- Click  PLACE

### 4.2 Place Unit by Clicking the Map

- Select faction and unit
- Click  CLICK MAP — button turns green, cursor becomes crosshair
- Click anywhere on the map — snaps to nearest town within ~5 km, otherwise places at field position

### 4.3 Add a Custom Town

- Click + ADD CUSTOM TOWN
- Type the town name and click OK
- Click the map to place — immediately available in all dropdowns

**Note:** Custom towns are session-only and must be re-added after reloading.

### 4.4 Map Overlay Opacity

Drag the PERIOD MAP slider in the PLACE tab to control how transparent the historical map overlay is. Set to 0 to hide it completely, 1.0 for full opacity.

### 4.5 Pre-Loaded Starting Positions — 16 APR 1809 at 0600

| Unit                                                                             | Starting Position | Faction               |
|----------------------------------------------------------------------------------|-------------------|-----------------------|
| Napoleon                                                                         | Donauwörth        | French — Army HQ      |
| Davout HQ, Morand, Saint-Hilaire, Gudin, Friant, Demont, Montbrun, Saint-Sulpice | Regensburg        | French — III Corps    |
| Vandamme HQ, Württemberg Cav, Württemberg Inf, Nansouty                          | Weihmichl         | French — VIII Corps   |
| Hiller                                                                           | Landshut          | Austrian — VI Corps   |
| Archduke Charles HQ, Hohenzollern, Rosenberg, Bellegarde, FML von Kienmayer      | Straubing         | Austrian — Hauptarmee |

## 5. MOVE Tab — Issuing Movement Orders

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### 5.1 Building a Route Order

- Select  ROAD mode (default) to use period road distances
- Select the unit from the dropdown
- Click + ADD WAYPOINT — a leg entry appears
- Pick destination town — distance, travel time, and ETA calculate automatically
- The route panel shows the exact road path: e.g. via Kelheim → Abensberg
- Enter depart time for Leg 1 (format: DD HHMM, e.g. 16 0800)
- Legs 2 and 3 auto-cascade — or enter a specific time to add rest
- Route summary shows total road distance, march time, and final ETA
- Click ⚡ ISSUE — route line appears on map following the road network

**Key difference from previous version:** ETAs now reflect road distances, not straight-line. Regensburg to Landshut is 70km by road (17.5 hrs at 4 km/hr) not the 51km straight line that would give 12.75 hrs. This matches your umpire's calculations.

### 5.2 Route Visualization

Issued routes draw on the map as dashed colored lines following the road network. Blue lines for French friendly units, red for enemy. Each waypoint shows a filled dot at the destination. As the clock advances units move along the road path — not in straight lines.

### 5.3 Canceling a Route

- Go to MOVE tab, select the unit
- Click ✕ CLEAR

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## 6. OOB Tab — Order of Battle

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Pre-loaded with the historical 1809 order of battle. Shows the full command hierarchy as a collapsible tree. Click any unit to select it on the map.

### 6.1 French OOB — Pre-loaded

Napoleon → Davout HQ (Morand, Saint-Hilaire, Gudin, Friant, Demont, Montbrun, Saint-Sulpice), Vandamme HQ (Württemberg Cav, Württemberg Inf, Nansouty), Massena HQ, Oudinot, Lannes, Lefebvre HQ (Wrede, Deroy), Berthier

### 6.2 Austrian OOB — Pre-loaded

Archduke Charles HQ → Hohenzollern, Rosenberg, Bellegarde, Ludwig, Kollowrat, Liechtenstein, FML von Kienmayer, Hiller, Klenau, AH Johann, AH Ferdinand

### 6.3 Editing

- ↺ RESET 1809 — restore full historical OOB with one click
- ✕ CLEAR ALL — remove all hierarchy
- ASSIGN PARENT — select unit and new parent, click ASSIGN

## 7. INTEL Tab — Contact Detection

| Alert                                                                                        | Trigger                         | Meaning                                    |
|----------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------|
|  ENGAGEMENT | Units within 5 km               | Forces in direct contact — battle imminent |
|  CONTACT    | Units within 25 km              | Forces in close proximity                  |
|  PROJECTED  | Route destinations within 20 km | Routes converging — future contact likely  |

The Movement Log records all actions with game-time timestamps. Last 30 entries shown.



## 8. Quick Reference

| I Want To...              | How                                                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Place a unit              | PLACE tab → faction/unit/town →  PLACE                                                                                                                |
| Place by clicking map     | PLACE tab → select unit →  CLICK MAP → click                                                                                                           |
| Add a custom town         | PLACE tab → + ADD CUSTOM TOWN → type name → click map                                                                                                                                                                                    |
| Issue a move order (road) | MOVE tab →  ROAD → select unit → + ADD WAYPOINT → town + time →  ISSUE |
| See road route detail     | Move order panel shows: "via Kelheim → Abensberg"                                                                                                                                                                                        |
| Check ETA                 | Shown automatically in waypoint panel and route summary                                                                                                                                                                                  |
| Switch perspective        | Click FRENCH or AUSTRIAN in perspective bar                                                                                                                                                                                              |
| Change map layer          | Header dropdown → Street / Game Map / Alps / Rhine-Danube                                                                                                                                                                                |
| Adjust map opacity        | PLACE tab → Period Map opacity slider                                                                                                                                                                                                    |
| View OOB hierarchy        | OOB tab                                                                                                                                                                                                                                  |
| Reset to 1809 OOB         | OOB tab →  RESET 1809                                                                                                                                   |
| See contact alerts        | INTEL tab                                                                                                                                                                                                                                |
| Remove a unit             | Select it → PLACE tab → × REMOVE SELECTED                                                                                                                                                                                                |
| Cancel a route            | MOVE tab → select unit → × CLEAR                                                                                                                                                                                                         |
| Run clock automatically   | Set speed →  PLAY                                                                                                                                     |

## 9. Troubleshooting

| Problem                             | Solution                                                                                                                                                                           |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Broken image icon on period map     | All four files must be in the same folder. Do not rename the JPG files.                                                                                                            |
| Map appears black                   | Check internet connection — Leaflet loads from Cloudflare CDN on first use.                                                                                                        |
| Road distance not calculating       | Verify both towns are connected in the road network. Try adding an intermediate town as a waypoint.                                                                                |
| Two towns show "no road connection" | Some small villages are not in the road network. Add an intermediate major town as a waypoint leg.                                                                                 |
| ETA seems too long                  | Check you are using  ROAD mode. Road distances are longer than straight-line — this is correct. |
| File security error in Chrome       | Rename file to remove spaces and parentheses. Use bavaria_cop_1809.html                                                                                                            |
| Unit not snapping to town           | Snap radius ~5 km. Zoom in and click closer to the gold dot.                                                                                                                       |

## 10. Credits and Sources

Built for the Flight of the Eagle — 1809 Danube Campaign wargame at the U.S. Army War College, Carlisle Barracks, Pennsylvania.

- John H. Gill, From Abensberg to Znaim: the Franco-Austrian War of 1809 (USAWC monograph, v.2, January 2026)
- Jay Luvaas, Napoleon on the Art of War (The Free Press, 1999)
- Napoleon's Historical Correspondence, Germany and Italy 1809
- Flight of the Eagle Umpire Reference V4
- Thiers/Dufour period maps: Histoire du Consulat et de l'Empire (Paulin, Paris) — public domain, georeferenced by Frank L. Adams III

Technical: Leaflet.js 1.9.4 | OpenStreetMap | Road network by Frank L. Adams III | Dijkstra shortest-path routing

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